# Kronberger P1_3: Diffusion with reflecting walls

We solve the diffusion equation

$$\frac{\partial p}{\partial t} = D\frac{\partial^2 p}{\partial x^2}$$

on the interval $[-L/2, +L/2]$ with reflecting (Neumann) boundary conditions

$$\frac{\partial p}{\partial x}\,|_{\pm L/2} = 0$$

and initial condition $p(x, 0) = \delta(x)$.

**Separation of variables.** We set $p(x, t) = X(x) \cdot T(t)$ and obtain

$$\frac{T'}{DT} = \frac{X''}{X} = -k^2.$$

This yields two ODEs:

$$T(t) = e^{-Dk^2 t}, \quad X(x) = A\cos(kx) + B\sin(kx).$$

**Boundary conditions.**

From $X'(\pm L/2) = 0$ we get

$$-Ak\sin(kL/2) + Bk\cos(kL/2) = 0 \quad (\text{at} + L/2)$$

$$+Ak\sin(kL/2) + Bk\cos(kL/2) = 0 \quad (\text{at} - L/2).$$

Adding: $2Bk\cos(kL/2) = 0 \Rightarrow B = 0$

Subtracting: $-2Ak\sin(kL/2) = 0 \Rightarrow \sin(kL/2) = 0 \Rightarrow k_n = \frac{2\pi n}{L}, \quad n = 1, 2, \ldots$

The case $k = 0$ gives the constant eigenfunction $X_0 = \text{const}$. Sine solutions are eliminated.

**General solution.**

$$p(x, t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi n x}{L}\right) \exp\left[-D\left(\frac{2\pi n}{L}\right)^2 t\right]$$

**Initial condition.** Using $p(x, 0) = \delta(x)$:

$$a_n = \frac{2}{L}\int_{-L/2}^{L/2} \delta(x)\cos\left(\frac{2\pi n x}{L}\right) \mathrm{d}x = \frac{2}{L}\cos(0) = \frac{2}{L}$$

and $a_0/2 = 1/L$ (normalization).

$$p(x, t) = \frac{1}{L} + \frac{2}{L}\sum_{n=1}^{\infty} \cos\left(\frac{2\pi n x}{L}\right) \exp\left[-D\left(\frac{2\pi n}{L}\right)^2 t\right]$$

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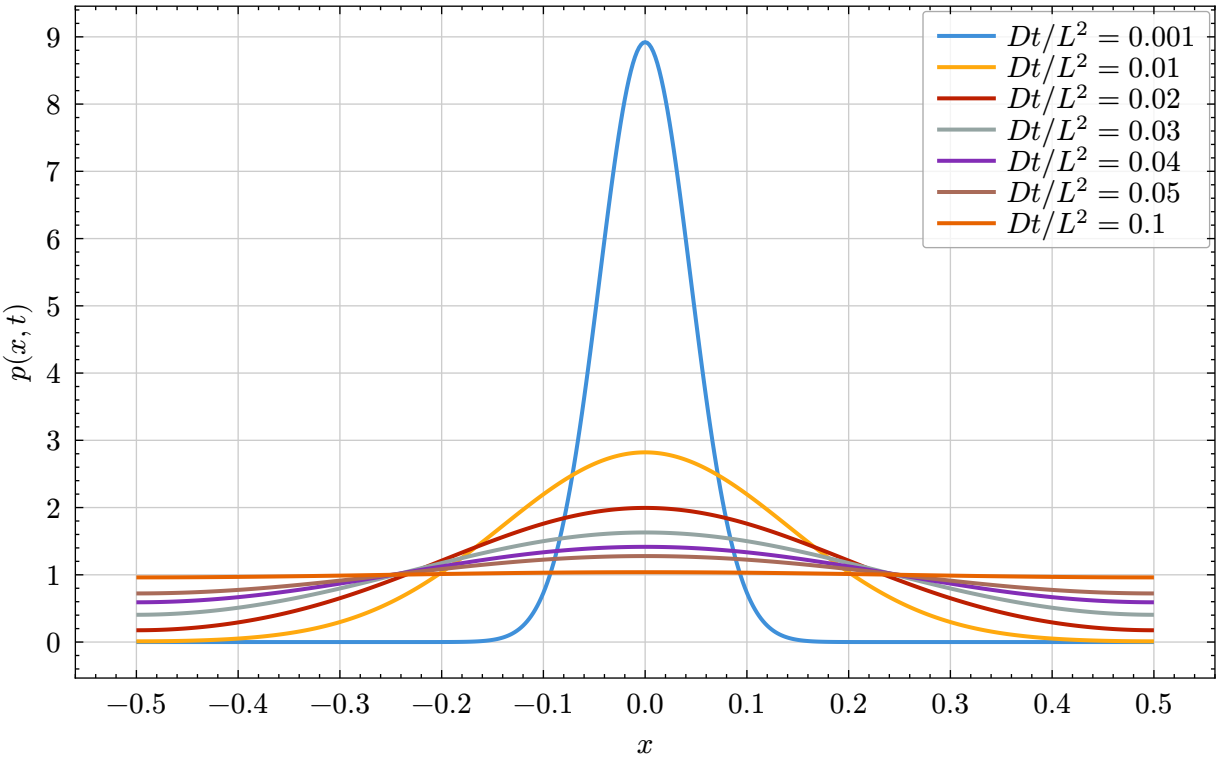


Figure 1: Analytical solution $p(x,t)$ for a diffusing particle confined to $[-L/2, +L/2]$ with reflecting walls, shown for various reduced times $Dt/L^2$.